

Desain Interaksi & Faktor Manusia

IF3151 Interaksi Manusia–Komputer

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Pengantar Desain Interaksi

Apa Itu Desain Interaksi?

Desain interaksi adalah disiplin yang berfokus pada perancangan produk dan layanan interaktif yang memfasilitasi interaksi antara manusia dengan sistem.

- Fokus pada perilaku dan pengalaman, bukan *hanya* estetika.
- Memadukan prinsip teknik, psikologi, dan desain.
- Memastikan sistem intuitif, efisien, dan menyenangkan digunakan.

Interaction design is about enabling people, not just operating machines.

Definisi & Evolusi

Interaksi desain berkembang dari UI tradisional menuju pengalaman yang memfasilitasi dialog antara manusia dan produk.

- John Kolko: “Desain interaksi adalah penciptaan dialog antara seseorang dan produk, layanan, atau sistem.”
- Menekankan hubungan fisik dan emosional serta konteks penggunaan.
- Arah evolusi: Mengintegrasikan komputasi yang meresap (*pervasive*) dan kecerdasan buatan.

Usability & UX

Usability

- Usability Goals (objektif, berorientasi pada kinerja):
 - *Effectiveness*: Bisakah pengguna mencapai tujuan mereka secara **akurat**?
 - *Efficiency*: Dapatkah mereka melakukannya dengan cepat dan **upaya minimal**?
 - *Safety*: Apakah kesalahan dapat **dicegah** atau **dipulihkan**?
 - *Utility*: Apakah sistem menyediakan fitur-fitur yang **dibutuhkan**?
 - *Learnability*: Apakah **mudah** bagi **pengguna baru** untuk mulai menggunakannya?
 - *Memorability*: Apakah **mudah digunakan kembali** setelah jeda waktu tertentu?
- Usability diukur berdasarkan seberapa mampu dan percaya diri perasaan pengguna.

UX

Pengalaman pengguna (UX) lebih luas daripada usability.

- Meliputi aspek emosi, estetika, dan nilai pribadi.
- Mencakup keseluruhan perjalanan pengguna sebelum, selama, dan sesudah interaksi.
- Mengintegrasikan brand, kepercayaan, dan harapan sosial.
- **Desirable aspects:** Fun, satisfying, engaging, aesthetically pleasing, emotionally fulfilling, etc.
- **Undesirable aspects:** Boring, frustrating, patronizing, gimmicky, annoying, etc.

Heuristik Usability Nielsen

in depth, next lecture ...

1. Visibility of system status
2. Match between system and real world
3. User control and freedom
4. Consistency and standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use
8. Aesthetic and minimalist design
9. Help users recognize, diagnose, and recover from errors
10. Help and documentation

Prinsip Dasar Interaksi

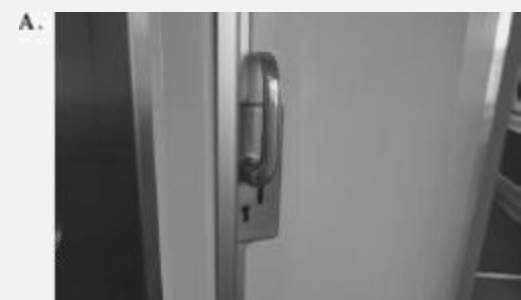
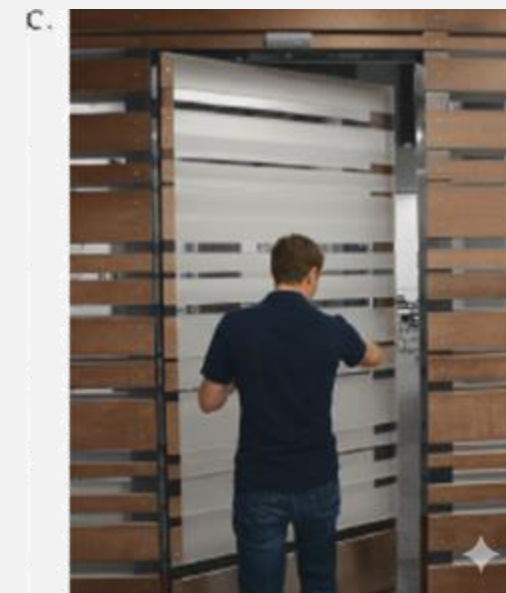
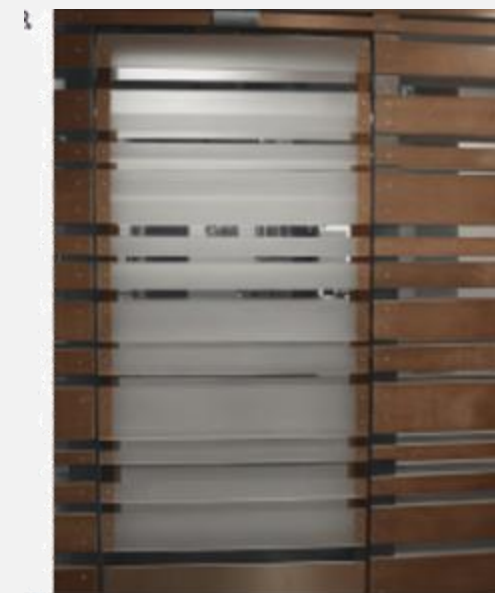
Affordances & Signifiers

Affordances

- Relationship between object properties + user capabilities.
- *Example:* A chair **affords sitting**; glass **affords seeing through** but **blocks passage** (anti-affordance).
- **Not a property**—it's relational (e.g., a heavy chair doesn't afford lifting for a child).
- Can be visible or not. If an affordance is invisible, we need ...

Signifiers

- **Signals** that communicate **where** and **how** to act.
- Can be intentional (labels, handles) or accidental (wear patterns, footprints).
- *Key insight:* **Signifiers > Affordances** in design—visibility matters most.
- Bad design = needing signs like “PUSH” on a door.



Mapping, Feedback, & Constraints

Mapping

- Relationship between controls and their effects
- **Natural mapping**: Spatial or cultural correspondence
- Poor mapping = confusion (e.g., stove knobs not aligned with burners)

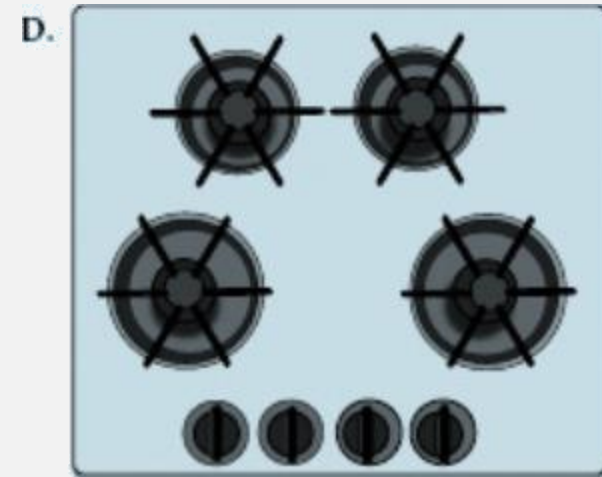
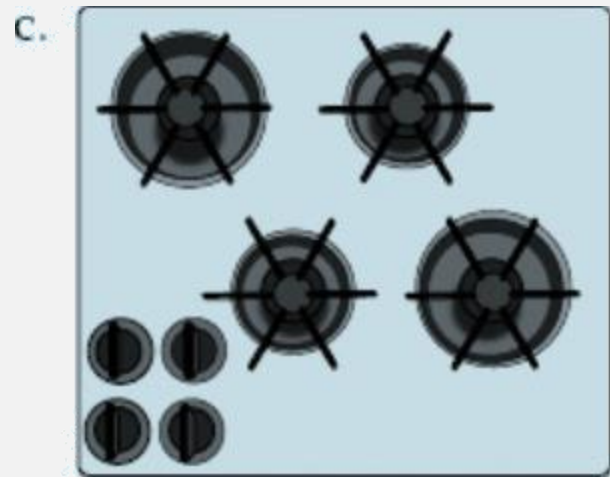
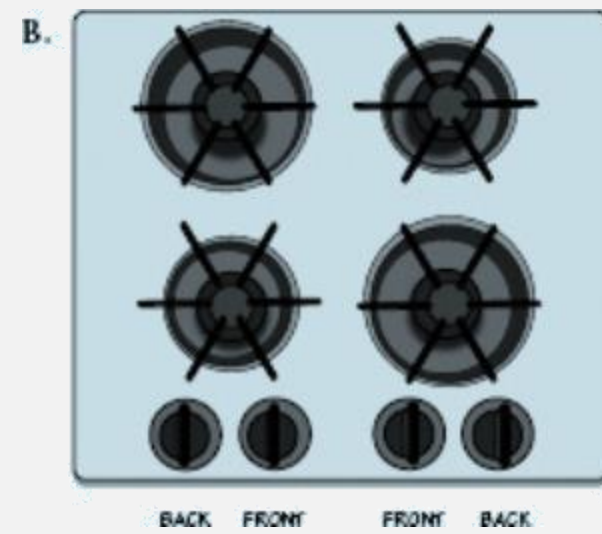
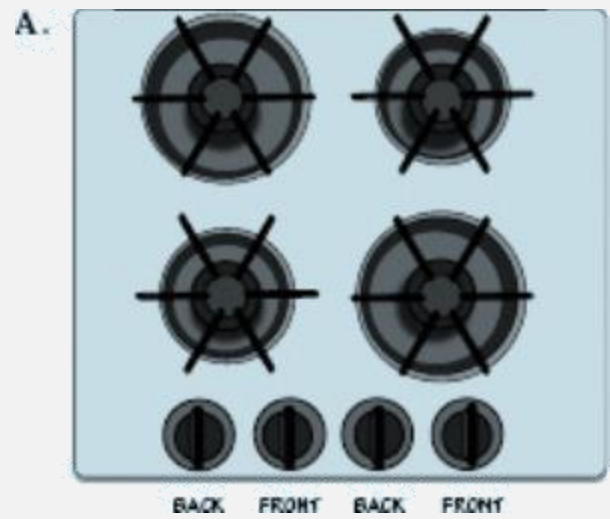
Feedback

- Immediate, clear response to user action
- Must be **informative**, **timely**, and **not overwhelming**
- *Bad feedback*: Elevator button gives no signal → people press repeatedly
- *Too much feedback*: Dishwasher beeping at 3 a.m.

Constraints

- Limit possible actions (physical, logical, cultural, semantic)
- Prevent errors (e.g., USB Type-A/B shape only fits one way)





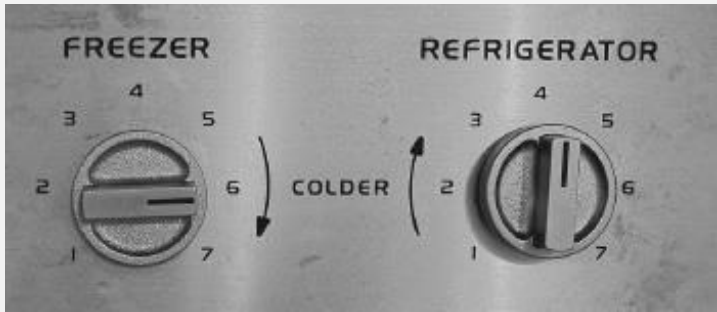
Conceptual Models

Conceptual Models

- User's mental explanation of how something works
- Built from **system image** (what the product communicates via design, manual, etc.)
- **Good model** → Predictable use, easy troubleshooting
- **Bad model** → Frustration (e.g., fridge with two interdependent controls labeled "Freezer"/"Fridge")

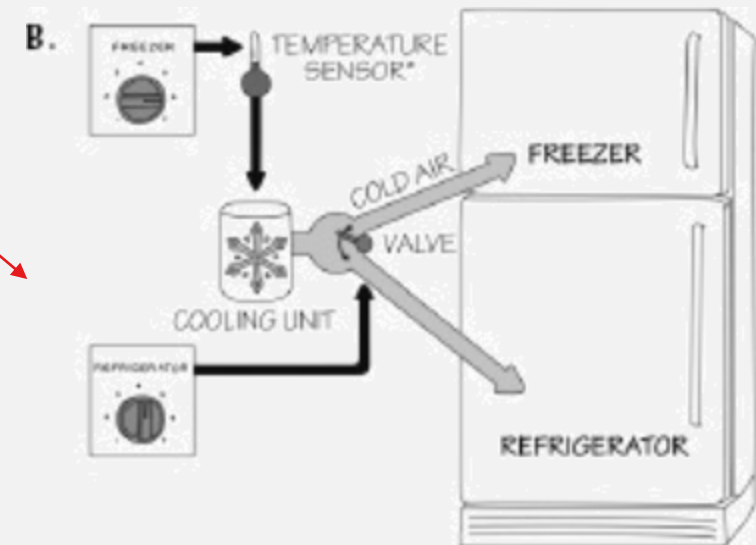
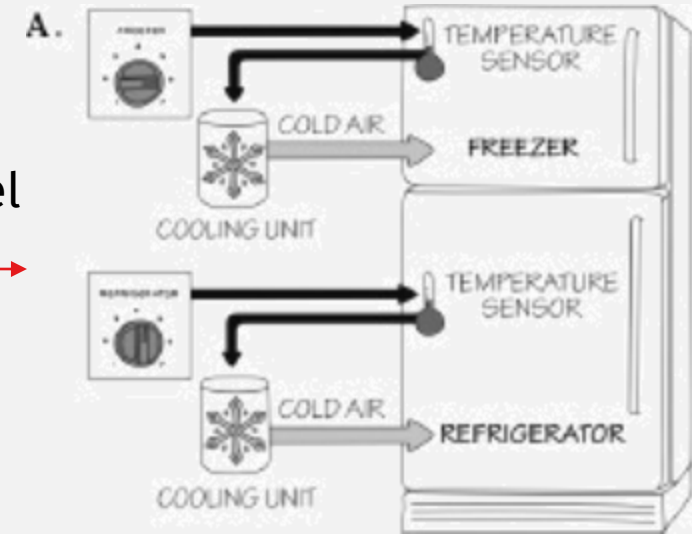


“Junghans Mega 1000 Digital Radio Controlled Watch. *There is no good conceptual model for understanding the operation of my watch. It has five buttons with no hints as to what each one does. And yes, the buttons do different things in their different modes. But it is a very nice-looking watch, and always has the exact time because it checks official radio time stations. (The top row of the display is the date: Wednesday, February 20, the eighth week of the year.)”*



user's conceptual model

designer's
actual model



Antarmuka & Gestalt

People see structure, not pixels — structure guides understanding.

Navigasi, Tata Letak, Hierarki

- Navigasi harus jelas dan konsisten – gunakan struktur menu yang logis dan mudah ditemukan.
- Tata letak memanfaatkan *grid* untuk menyusun elemen; kedekatan fisik menyiratkan hubungan.
- Hierarki informasi dibangun melalui ukuran, warna, dan posisi untuk memandu perhatian.
- Gunakan *whitespace* untuk meningkatkan keterbacaan dan mengurangi beban kognitif.

■ **The more important something is, the more prominent it is.** The most important elements are either larger, bolder, in a distinctive color, set off by more white space, or nearer the top of the page—or some combination of the above.

Very important

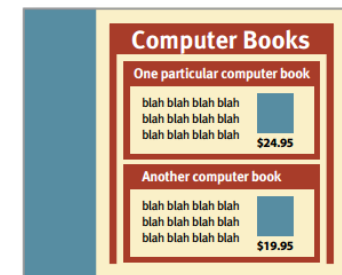
A little less important

Nowhere near as important

■ **Things that are related logically are related visually.** For instance, you can show that things are similar by grouping them together under a heading, displaying them in the same visual style, or putting them all in a clearly defined area.

Books
Music
Movies
Games

■ **Things are “nested” visually to show what’s part of what.** For instance, a site section name (“Computer Books”) would appear above the titles of the individual books, reflecting the fact that the books are part of the section. And each book title in turn would span all the elements that make up the description of that book.

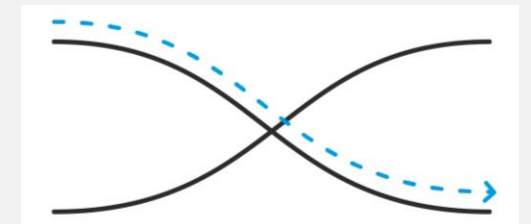
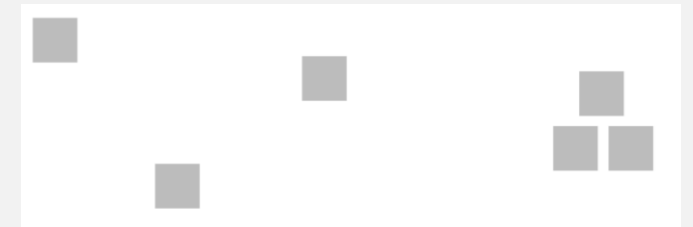
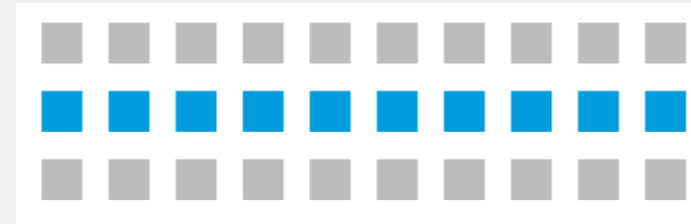


Isyarat Visual & Gestalt

Prinsip Gestalt menjelaskan bagaimana pengguna mengelompokkan dan menafsirkan elemen visual:

- *Similarity* – elemen serupa dipersepsikan sebagai kelompok.
- *Proximity* – elemen yang dekat dipersepsikan satu kelompok.
- *Closure* – otak melengkapi bentuk yang tidak lengkap.
- *Continuity* – perhatian mengikuti garis yang halus.
- dst.

Gunakan prinsip ini untuk mengarahkan fokus dan membentuk struktur visual.



Model Mental & Kognisi

The Two Gulfs

- **Gulf of Execution:** Difficulty figuring out *how to use* something
 - Bridged by: signifiers, constraints, mappings, conceptual models
- **Gulf of Evaluation:** Difficulty understanding *what happened* after acting
 - Bridged by: feedback + good conceptual model



The Seven Stages of Action

(Framework for understanding human behavior with technology)

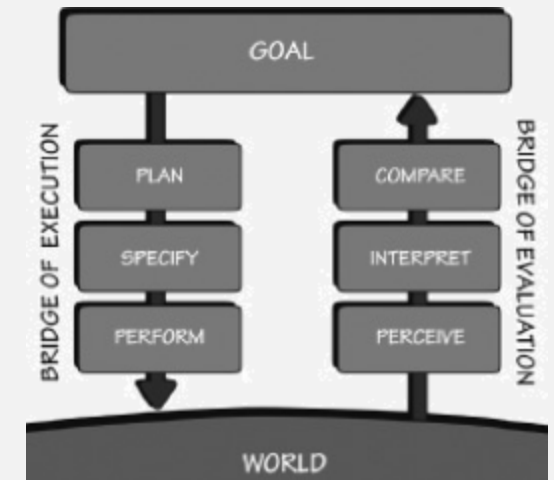
Execution (Doing)

1. **Goal:** Form intention (e.g., “I need more light”)
2. **Plan:** Choose action sequence (e.g., “Turn on lamp”)
3. **Specify:** Determine exact steps (e.g., “Reach right hand to switch”)
4. **Perform:** Carry out action

Evaluation (Interpreting)

5. **Perceive:** Observe system state (e.g., “Is the light on?”)
6. **Interpret:** Make sense of perception
7. **Compare:** Check if outcome matches goal

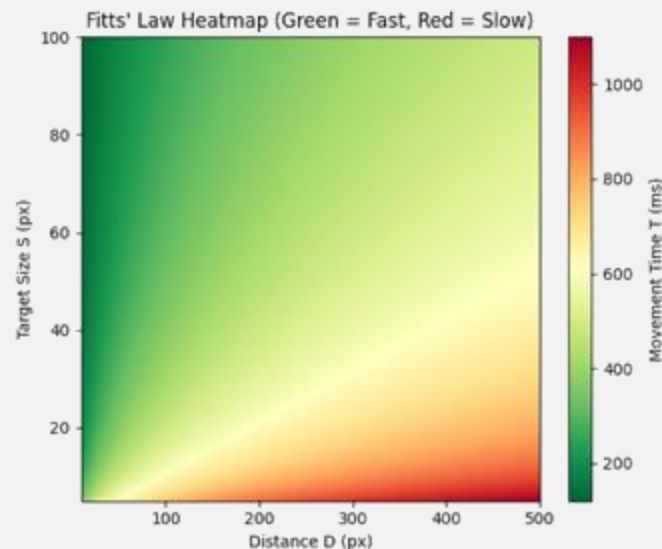
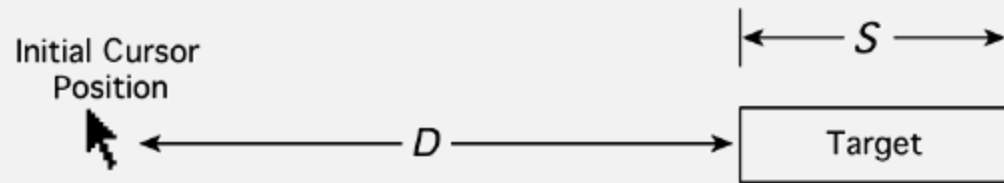
Most stages are **subconscious** during routine tasks; only surface during errors or novelty.



Persepsi & Motorik

Fitt's Law

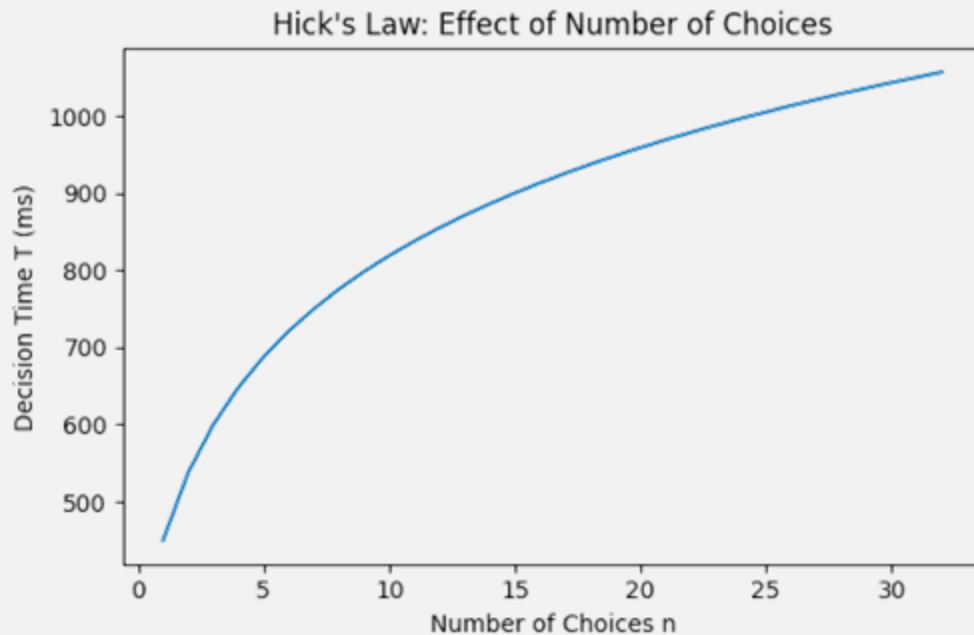
$$\text{Time (in msec)} = a + b \log_2 (D / S + 1)$$



- Hukum Fitts menyatakan bahwa waktu untuk mencapai target meningkat dengan jarak dan berkurang dengan ukuran target.
- Desain:
 - buat target besar & dekat;
 - gunakan tepi layar untuk target “tak terbatas”;
 - minimalkan perjalanan pointer.

Hick's Law

$$\text{Time (in msec)} = a + b \log_2 (n + 1)$$



- Hukum Hick menyatakan bahwa semakin banyak pilihan dan semakin kompleks pilihan, semakin lama waktu yang dibutuhkan untuk memutuskan.
- Desain:
 - batasi jumlah opsi,
 - pecah tugas kompleks menjadi langkah kecil,
 - beri highlight opsi rekomendasi,
 - gunakan progressive disclosure.

Proses Desain Interaksi

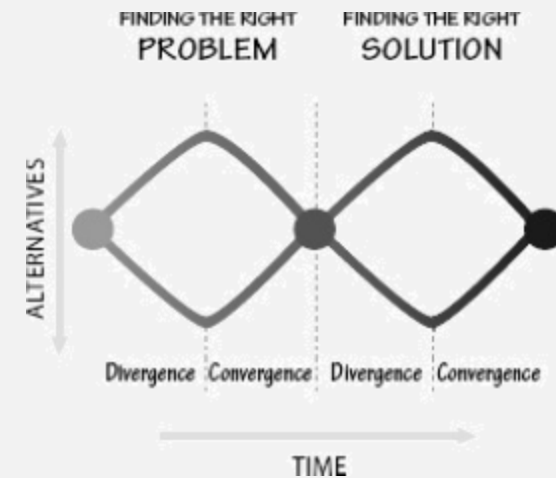
Filosofi *Human-Centered Design* (HCD)

- **Premise:** Technology should serve people, not the reverse.
- **Mindset Shift:**
 - ✗ “User error” → ✓ “Design failure”
 - ✗ Blame the person → ✓ Fix the system
- **Goal:** Create products that are usable, understandable, enjoyable, and resilient to error.

The Double-Diamond Model (British Design Council)

- **First Diamond: Find the Right Problem**
 - **Diverge:** Explore broadly—observe, question assumptions, uncover hidden needs.
 - **Converge:** Define the core, underlying problem.
- **Second Diamond: Find the Right Solution**
 - **Diverge:** Generate many ideas (ideation), encourage wild concepts.
 - **Converge:** Select and refine the best solution.

Both phases require time, iteration, and tolerance for ambiguity.



Human-Centered Design (HCD) Process

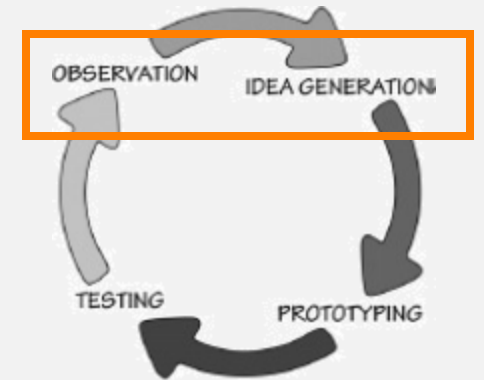
(A four-stage iterative cycle within the double-diamond)

1. Observation

- Study people in their natural context (applied ethnography).
- Focus on *activities*, not just demographics.
- Understand goals, frustrations, and workarounds.

2. Idea Generation (Ideation)

- Brainstorm without judgment.
- Encourage “stupid questions” that challenge assumptions.
- Quantity over early quality.



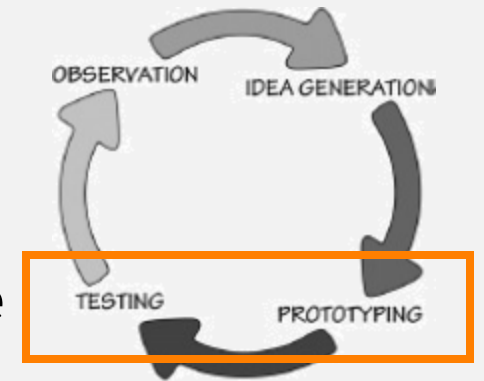
Human-Centered Design (HCD) Process

3. Prototyping

- Build quick, low-fidelity mockups (sketches, cardboard, role-play).
- *Wizard of Oz*: Simulate complex systems with human “wizards” behind the scenes.
- Test concepts before investing in full development.

4. Testing

- Observe target users interacting with prototypes.
- Use pairs: one operates, one narrates thoughts aloud.
- Test early, test often—*5 users per iteration* is often sufficient.



Iterate: Cycle through all four stages repeatedly until the solution fits.

Tasks vs. Activities

- **Activity:** High-level goal (e.g., “enjoy music”).
- **Task:** Component action (e.g., “buy song,” “create playlist”).

Great design supports the whole activity, not just isolated tasks.

- *Apple’s iPod success:* Integrated hardware, software, store, and ecosystem for the *activity* of music enjoyment.

Human Error & Toleransi

Two Fundamental Types of Error

- **SLIPS:** Action is correct, but execution fails.
 - *Example:* Pouring milk into coffee, then putting the coffee cup in the fridge.
 - Occur at **lower (subconscious) levels** of the action cycle.
- **MISTAKES:** Goal or plan is wrong.
 - *Example:* Using pounds instead of kilograms for fuel (Boeing 767 “Gimli Glider”).
 - Occur at **higher (conscious) levels** of cognition.





Why Errors Happen: Systemic Causes

- **Unnatural task demands:** Require sustained attention, precision, multitasking.
- **Interruptions:** Break action cycles; memory fails to resume correctly.
- **Time pressure:** Hurry leads to skipped steps or misjudgment.
- **Poor feedback:** No way to verify actions or detect errors.
- **Social & institutional pressures:** Don't delay flight, don't shut down plant, etc.
 - Case: Tenerife airport disaster (1977) — KLM took off without clearance due to time/weather pressure.

Design Strategies to Prevent & Mitigate Error

- **Add constraints:** Physical, logical, semantic, cultural (see Ch. 4).
- **Use forcing functions:**
 - *Interlocks:* Sequence enforced (microwave won't run if door open).
 - *Lock-ins:* Keep system active until task done (save-before-quit dialog).
 - *Lockouts:* Prevent entry/action (childproof caps, stairwell gates).
- **Provide Undo:** Reverse actions easily; multiple levels preferred.
- **Sensibility checks:** Reject implausible inputs (e.g., radiation dose 1000× normal).
- **Minimize mode errors:** Eliminate modes if possible; make active mode highly visible.
- **Support interruptions:** Preserve context so users can resume easily.

Design Principles for Error-Tolerant Systems

- **Assume error will happen**—design for resilience.
- **Put knowledge in the world**, not just in the head.
- **Bridge the Gulfs:**
 - *Execution*: Use signifiers, constraints, natural mappings.
 - *Evaluation*: Provide clear, immediate feedback.
- **Replace blame with curiosity**: “What led to this?” → “How can we prevent it?”